LLM Post-training Data Selection

A controlled study of budget-equivalent selection and candidate-utility reliability

RESEARCH QUESTION

Under matched sample count, response-supervision tokens, and data composition, does targeted instruction selection outperform matched random selection?

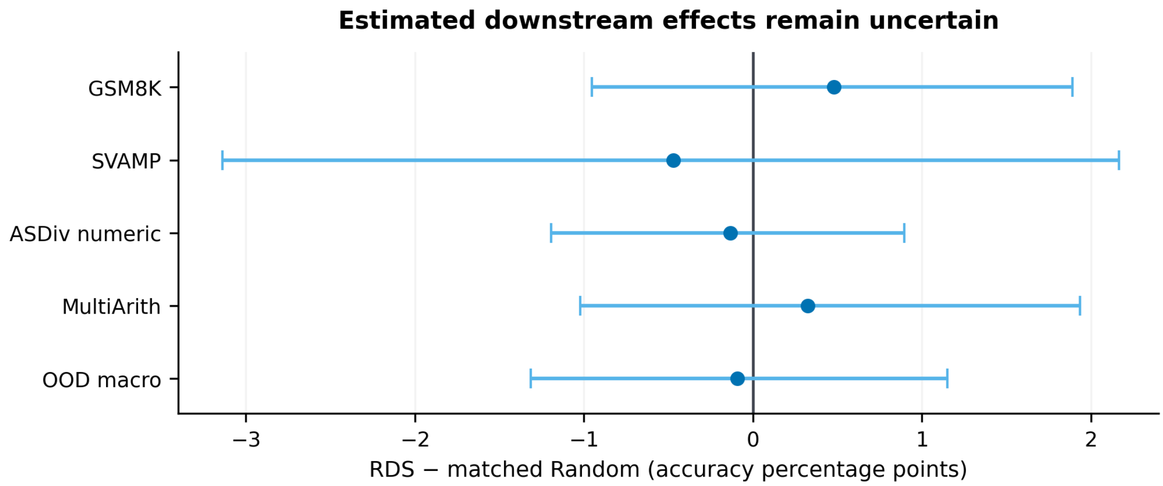
Controlled design

2 selection methods	**4** list realizations per method	**3** training seeds	**24** audited cells

MATCHED FACTORS

- 500 examples, response-supervision exposure, source x answer-length composition
- same base model, LoRA recipe, optimizer protocol, parser, and evaluation

Main result



Point = estimated difference; horizontal bar = 95% interval

BOUNDED INTERPRETATION

GSM8K: +0.480 pp

95% interval [-0.954, +1.889] pp

OOD macro: -0.094 pp

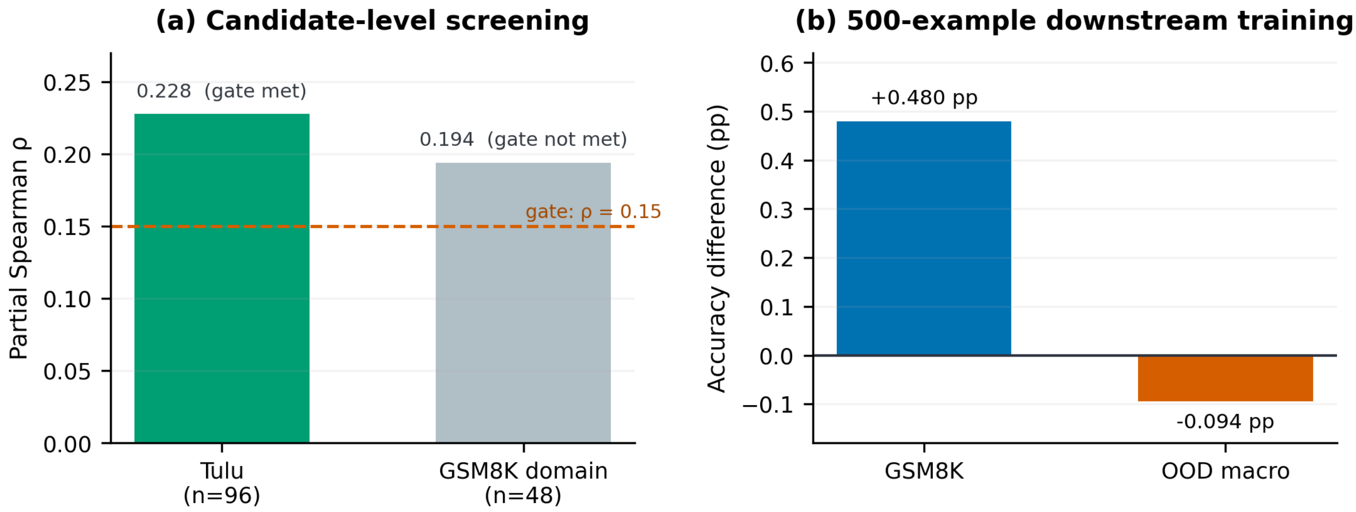
95% interval [-1.316, +1.149] pp

- No reliable downstream advantage was observed for the frozen error-conditioned selection policy.
- The intervals still include relevant positive and negative effects.
- The study therefore establishes neither ineffectiveness nor equivalence.

From candidate signal to a falsifiable next study

Why a local screening result did not become a stable set-level benefit

Candidate-level signal did not yield a reliable set-level gain

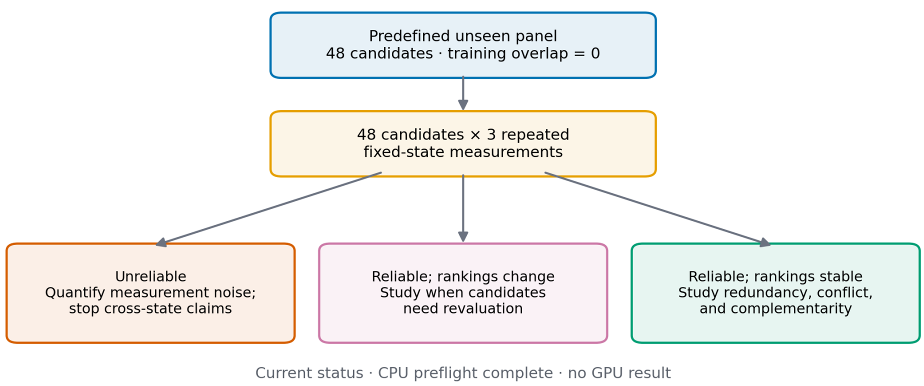


OBSERVED GAP

Tulu96 met the preregistered screen (partial Spearman 0.228 >= 0.15; one-sided p = 0.073 <= 0.10; positive top-minus-bottom utility), but the 500-example downstream study did not show a stable gain.

Next validation step

Fixed-state reliability follow-up: measure before cross-state comparison



FIXED-STATE RELIABILITY

- 48 predefined candidates
- training overlap = 0
- 48 × 3 repeated measures
- CPU preflight complete
- no GPU result

CROSS-STATE TESTING IS GATED

Four-week research module

WEEK 1

Qualify fresh-process probes and seed semantics.

WEEK 2

Run and audit 48 × 3 fixed-state reliability.

WEEK 3

Apply the predefined stop/go decision before any cross-state evaluation.

WEEK 4

Deliver evidence, bounded claims, and next protocol.